

Stacked Solar-Shade Radiator Architecture

Variable Reference — Rev B

Variable Reference | Rev B — September 2026

1. Symbols

Symbol	Name	Definition	Nominal
ε_{rad}	Radiator emissivity	Fraction of blackbody emission from the radiator coating. Clean white coatings (AZ-93, Z-93P) 0.85–0.92; 21-day dust exposure 0.70–0.80.	0.88 clean / 0.75 dusty
α_{s}	Radiator solar absorptance	Fraction of incident sunlight absorbed by the radiator coating. Clean 0.09–0.15; dusty 0.20–0.30.	0.12 / 0.25
T_{rad}	Radiator temperature	Panel surface temperature set by the fluid loop outlet.	310 K
S	Solar constant	Irradiance at 1 AU.	1,361 W/m ²
α	Sun elevation	Angle of the sun above the local horizon. Polar sites 1.5–15°; noon elevation at a fixed site is 90° minus latitude. The sun moves about 0.5° per hour.	site-dependent
σ	Stefan–Boltzmann constant		5.67×10^{-8} W/m ² K ⁴
T_{sink}	Sky sink temperature	Deep space; negligible in the balance.	~4 K
h	Canopy height	Clearance from the radiator face to the canopy underside.	1–4 m (fixed assets)
F	View factor, radiator to canopy	Fraction of the radiator's hemisphere occupied by the canopy underside. Computed by numerical integration over both rectangles; equals the closed form for equal parallel plates. Rises with overhang and falls with height.	0.25–0.99 (see tables)
$\varepsilon_{\text{back}}$	Canopy backside emissivity	Underside of the array. Bare 0.75–0.85; vacuum-deposited aluminium 0.05–0.20.	0.1 coated / 0.82 bare
T_{array}	Canopy backside temperature	Tracks the cell temperature; 300 K at grazing sun, 330 K nominal, 370–400 K at equatorial noon.	330 K
α_{shade}	Shade design elevation	Lowest sun elevation at which the canopy must fully shade the radiator; sets the overhang.	noon elevation – 20°
overhang	Canopy overhang	Extension of the canopy beyond the radiator on each sun-facing edge: $h / \tan \alpha_{\text{shade}}$.	see tables
η	Photovoltaic efficiency	Canopy array conversion efficiency.	0.28

2. Equations

Projected sunlight on a horizontal surface: $q_{\text{sun}} = S \sin \alpha$

Absorbed by the radiator: $q_{\text{abs}} = \alpha_s S \sin \alpha$

Unshaded net rejection: $q_{\text{unshaded}} = \varepsilon_{\text{rad}} \sigma T_{\text{rad}}^4 - \alpha_s S \sin \alpha$

Canopy backradiation onto the radiator: $q_{\text{back}} = F \varepsilon_{\text{back}} \sigma T_{\text{array}}^4$

Shaded net rejection: $q_{\text{shaded}} = \varepsilon_{\text{rad}} \sigma T_{\text{rad}}^4 (1 - F) - q_{\text{back}}$

Overhang per sun-facing edge: $d = h / \tan \alpha_{\text{shade}}$

View factor, equal parallel rectangles $a \times b$ at separation c ($X = a/c$, $Y = b/c$,

$A = \sqrt{1+X^2}$, $B = \sqrt{1+Y^2}$):

$F = 2/(\pi XY) \cdot [\frac{1}{2} \ln(A^2 B^2 / (1+X^2+Y^2)) + XB \operatorname{atan}(X/B) + YA \operatorname{atan}(Y/A) - X \operatorname{atan} X - Y \operatorname{atan} Y]$

General case: $F = (1/A_1) \iint \cos \theta_1 \cos \theta_2 / (\pi r^2) dA_1 dA_2$, integrated numerically

Canopy power per unit area: $p = \eta S \sin \alpha$ (horizontal canopy)

3. Reference values used in the Rev B tables

Quantity	Value
Radiator for view-factor tables	2 × 3 m, horizontal
Canopy for the net-rejection table	radiator + 1 m overhang all round, heights 0.4–3.0 m
Canopy for the site table	sized to shade at noon elevation – 20°, heights 1–4 m
Equatorial sizing point	3 m height, shade from 70°, overhang 1.1 m, F 0.40
Masses	radiator 8–12 kg/m ² with loop; array 4–8 kg/m ² ; truss 1.5 kg/m ² per m of height; posts 10 kg each
Polar comparison	6° sun: unshaded 444 / 357 W/m ² ; roof stack at 0.4 m 74 W/m ²

Rev B, September 2026. Replaces the February 2026 variable reference.